
HOMEWORK DAY 12 – *Characteristic eqn. Real roots (distinct and repeated) §2.2.1, 2.2.2*

1. In each of these problems, find the general solution of the given differential equation

(a) $x'' + 2x' - 3x = 0$

(b) $6x'' - x' - x = 0$

(c) $x'' + 5x' = 0$

(d) $x'' - 9x' + 9x = 0$

2. Find the solution of the given initial value problem. Sketch the graph of the solution and describe its behaviour as t increases.

(a) Logan, p90, # 3

(b) $x'' + 4x' + 3x = 0$, $x(0) = 2$, $x'(0) = -1$

(c) $4x'' - x = 0$, $x(0) = 1$, $x'(0) = -1$

(d) $6x'' - 5x' + x = 0$, $x(0) = 4$, $x'(0) = 4$

HOMEWORK DAY 13 – *Repeated Roots. Complex numbers.*

3. Solve the following initial value problem and sketch a graph of the solution.

(a) $x'' + 4x' + 4x = 0$, $x(0) = 2$, $x'(0) = 1$

(b) $9y'' - 12y' + 4y = 0$, $y(0) = 2$, $y'(0) = -1$

4. p90: 4.

5. p90: 6.

6. p90: 7.

7. Use Euler's formula to write the given expression in the form $a + ib$:

(a) e^{1+2i}

(b) e^{2-3i}

(c) $e^{i\pi}$

(d) $e^{2-(\pi/2)i}$

8. Using Euler's formula, show that

(a) $\cos t = (e^{it} + e^{-it})/2$

(b) $\sin t = (e^{it} - e^{-it})/(2i)$.

HOMEWORK DAY 14 – *Complex roots. §2.2.3*

9. p94: 1a

p94: 1b

p94: 1c

p94: 1d

p94: 1e

p94: 1f

10. Solve the following initial value problems.

(a) $x'' + 4x = 0$, $x(0) = 0$, $x'(0) = 1$

(b) $x'' + 4x' + 5x = 0$, $x(0) = 1$, $x'(0) = 0$

(c) $y'' - y' + 9.25y = 0$, $y(0) = 3$, $y'(0) = 1$

11. p94: 3

12. p94: 5